

Problem 9.11

A bank credits interest on deposits quarterly at a rate of 2% per quarter. What is the nominal interest rate per annum for interest compounded quarterly? Find the annual effective rate of interest.

Solution

The bank credits interest quarterly at a rate of 2% per quarter. Therefore, the quarterly effective interest rate is

$$\frac{i^{(4)}}{4} = 0.02.$$

Thus, the nominal annual interest rate compounded quarterly is

$$i^{(4)} = 4(0.02) = 0.08.$$

So,

$$\boxed{i^{(4)} = 8\%}.$$

To find the annual effective interest rate, use the relationship

$$1 + i = \left(1 + \frac{i^{(4)}}{4}\right)^4.$$

Since $i^{(4)}/4 = 0.02$, we have

$$1 + i = (1.02)^4.$$

Therefore,

$$i = (1.02)^4 - 1.$$

Now compute:

$$i \approx 0.08243216.$$

Thus, the annual effective rate of interest is

$$i \approx 8.2432\%.$$